

MINYA, Egypt

WMO: 623870

Lat: 28.083N

Lon: 30.733E

Elev: 37

StdP: 100.88

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 4.5	(c) 5.6	(d) -1.1	(e) 3.5	(f) 12.1	(g) 0.2	(h) 3.9	(i) 12.5	(j) 9.2	(k) 17.2	(l) 8.1	(m) 16.2	(n) 1.1	(o) 340	(p) 0.405

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 12.7	(c) 40.2	(d) 22.0	(e) 38.7	(f) 21.9	(g) 37.4	(h) 21.8	(i) 24.6	(j) 34.2	(k) 24.0	(l) 33.7	(m) 23.4	(n) 33.3	(o) 3.2	(p) 20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.1	16.9	28.2	21.2	15.9	28.1	20.4	15.2	28.0	74.6	34.0	72.1	33.5	69.9	33.3	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.3	(b) 7.4	(c) 6.6	(e) 2.6	(f) 43.9	(g) 1.5	(h) 1.8	(i) 1.5	(j) 45.2	(k) 0.6	(l) 46.3	(m) -0.2	(n) 47.4	(o) -1.2	(p) 48.7		
(a) 8.3	(b) 7.4	(c) 6.6	(e) 1.3	(f) 26.5	(g) 1.5	(h) 1.5	(i) 0.2	(j) 27.6	(k) -0.7	(l) 28.5	(m) -1.5	(n) 29.3	(o) -2.6	(p) 30.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.6	13.1	14.8	18.1	22.3	26.8	29.3	29.9	30.1	28.6	24.6	19.1	14.6
	DBStd	6.67	1.98	2.76	3.20	3.40	2.98	2.20	1.59	1.50	1.85	2.62	2.59	2.14
	HDD10.0	2	1	1	0	0	0	0	0	0	0	0	0	0
	HDD18.3	452	164	104	43	3	0	0	0	0	0	0	20	117
	CDD10.0	4616	96	135	251	369	522	579	617	623	558	451	272	143
	CDD18.3	2026	1	6	36	123	263	329	358	365	308	193	42	2
	CDH23.3	25808	14	87	438	1457	3375	4516	4983	5028	3769	1788	328	24
CDH26.7	14180	1	22	155	686	1895	2679	2921	2934	2061	751	73	3	
Wind	WSAvg	2.7	2.1	2.3	2.8	3.2	3.5	3.6	2.7	2.6	3.2	2.7	2.2	2.0
Precipitation	PrecAvg	3	0	1	1	0	0	0	0	0	0	0	0	0
	PrecMax	19	6	19	10	2	1	0	0	0	0	0	9	4
	PrecMin	0	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	5	1	4	2	0	0	0	0	0	0	0	2	1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.1	29.6	34.6	39.3	42.4	42.6	40.9	40.2	40.1	36.8	31.6	26.1
		MCWB	15.9	18.2	19.8	21.1	20.7	22.0	23.2	23.3	22.2	21.7	19.4	17.0
	2%	DB	22.3	25.7	30.6	36.1	39.0	40.0	38.7	38.6	38.0	34.2	28.6	23.2
		MCWB	14.4	15.6	17.8	19.6	19.9	21.5	22.9	23.1	21.7	21.0	19.0	15.5
	5%	DB	20.5	23.4	27.8	33.2	36.9	38.0	37.4	37.3	36.2	32.4	26.7	21.6
		MCWB	13.4	14.5	16.6	18.5	19.6	21.3	22.6	22.8	21.6	20.6	18.0	14.7
	10%	DB	19.1	21.4	25.4	30.6	34.8	36.2	36.2	36.2	34.7	30.7	25.1	20.3
		MCWB	12.6	13.6	15.5	17.6	19.1	20.9	22.2	22.5	21.3	20.1	17.3	14.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.0	18.9	20.9	22.3	22.1	23.9	25.3	25.7	24.2	23.6	21.3	18.4
		MCDB	23.7	28.6	32.1	37.3	36.9	36.6	35.7	34.3	33.7	32.3	28.0	23.7
	2%	WB	15.1	16.3	18.7	20.4	21.1	23.0	24.4	24.7	23.4	22.4	20.0	16.4
		MCDB	21.0	24.0	28.6	33.8	35.5	35.6	34.6	33.8	33.2	30.8	26.7	21.9
	5%	WB	13.9	15.0	17.2	19.1	20.3	22.3	23.7	24.1	22.8	21.5	18.9	15.2
		MCDB	19.5	22.2	26.6	31.5	34.1	34.7	33.9	33.3	32.5	29.7	25.1	20.7
	10%	WB	13.0	13.9	15.9	18.0	19.7	21.6	23.1	23.6	22.2	20.7	17.9	14.4
		MCDB	18.6	20.7	24.5	29.3	33.0	33.8	33.3	32.8	31.6	28.7	23.8	19.6
Mean Daily Temperature Range	5% DB	MDBR	12.9	13.3	13.8	15.1	14.9	14.0	13.1	12.7	12.6	11.9	12.3	12.3
		MCDBR	14.7	15.9	17.0	19.3	18.0	16.4	14.6	13.8	14.9	14.1	13.8	13.6
		MCWBR	8.4	7.8	7.7	7.3	5.4	4.6	4.5	4.3	4.3	4.8	6.3	7.2
	5% WB	MCDBR	14.2	14.8	16.3	18.0	16.1	14.6	13.6	12.7	13.2	12.4	12.7	12.9
		MCWBR	8.5	7.7	7.9	7.5	5.4	4.7	4.6	4.4	4.4	4.8	6.4	7.3
Clear-Sky Solar Irradiance	taub	0.403	0.438	0.484	0.561	0.574	0.494	0.481	0.473	0.481	0.481	0.419	0.401	
	taud	2.040	1.938	1.820	1.670	1.673	1.874	1.928	1.955	1.923	1.905	2.034	2.057	
	Ebn at Noon	815	819	807	758	748	806	816	821	799	766	791	795	
	Edh at Noon	145	174	208	250	249	203	192	185	185	178	145	137	
All-Sky Solar Radiation	RadAvg	4.13	5.04	6.32	7.24	7.83	8.28	8.07	7.54	6.70	5.46	4.40	3.84	
	RadStd	0.10	0.20	0.18	0.15	0.18	0.09	0.08	0.07	0.13	0.10	0.10	0.08	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.57	+0.75	N/A	N/A	N/A	+0.67	N/A	N/A	N/A
(47) Regional (2 neighbors)		+0.44	+0.46	+0.68	N/A	+0.50	+0.85	N/A	N/A	N/A

Nomenclature: See separate page